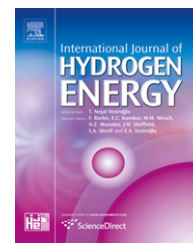


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Comparative study of eleven equations of state in predicting the thermodynamic properties of hydrogen

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ABSTRACT

Eleven equations of state are employed to predict the vapor pressures, liquid and vapor densities, liquid and vapor heat capacities, and vaporization enthalpies and entropies of normal hydrogen along the coexistence curve. The volumetric and thermal properties of gaseous hydrogen together with the speeds of sound, Joule–Thomson coefficients and inversion curves for wide ranges of temperature and pressure are predicted as well. The results are compared with experimental data and the recommended values of standard thermodynamic tables. The best equations of state in predicting the properties of hydrogen (saturated and supercritical) are introduced and reported.

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1. Introduction

Hydrogen has found applications in fuel cells, food industry, metal production, pharmaceuticals, aerospace, electronics and power generation. Hydrogen has been produced primarily using steam reforming of natural gases and oil mixtures [1–3]. Recent investigations have been focused on the reforming of ethanol [4] and glycerol [5] as well.

Accurate predictions of hydrogen properties are important for product and process design. Equally important are the equations of state (EoSs) that predict the properties of hydrogen and mixtures containing hydrogen. Hydrogen is a low molecular weight substance. At low temperatures and

high densities, hydrogen exhibits quantum behavior. It means that in certain applications, e.g. cryogenics, EoSs must appropriately be modified to take into account the impact of the quantum behavior [6]. However, remote from this region, where most applications of hydrogen are carried out, hydrogen shows classical behavior.

Hydrogen properties were predicted/correlated using theoretical EoSs [7,8]. Ibrahimoglu et al. [9] analyzed pressure–density and temperature–volume diagrams for hydrogen. Appropriate EoSs for predicting hydrogen properties were also developed, e.g. see Ref. [10]. Moreover, molecular simulation techniques for predicting hydrogen solubility in hydrocarbons, where hydrogen finds applications in hydrogenation,

Abbreviations: %AAD, Percent average absolute deviation; BWR, Benedict–Web–Rubin; NB, Nasrifar–Bolland; PR78, Peng–Robinson developed in 1978; PRGGPR, Peng–Robinson–Gasem–Gao–Pan–Robinson; PRMC, Peng–Robinson–Mathias–Copeman; PRSV, Peng–Robinson–Stryjek–Vera; PRTCC, Peng–Robinson–Twu–Coon–Cunningham; PSRK, Predictive Soave–Redlich–Kwong; PT, Patel–Teja; PTV, Patel–Teja–Valderrama; RK, Redlich–Kwong; RKS, Redlich–Kwong–Soave; RKMC, Redlich–Kwong–Mathias–Copeman; RKSU, Redlich–Kwong–Square–Well; SAFT-VR, Statistical associating fluid theory with variable range; SW, Square–Well.

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coal liquefaction and hydro treatment processes, were performed successfully [11]. Hydrogen solubility in liquid hydrocarbons was also calculated successfully using cubic EoSs [12,13]. In fact, cubic EoSs are conveniently used in engineering applications. Yang [14] used a multi parameter EoS fitted to experimental data in thermodynamic analysis of refueling of a hydrogen tank. Hadj-Kali et al. [15] used the cubic Peng–Robinson (PR) [16] EoS to evaluate the phase equilibria of HI_x (H_2O – HI – I_2) system. The system HI_x was also studied by Paricaud et al. [17] using statistical associating fluid theory with variable range (SAFT–VR) [18]. Mohamed and Paraschivoiu [19] used the Beattie–Bridgeman EoS [20] to assess the high pressure behavior of hydrogen when released from a chamber. Schouten et al. [21] used the PR EoS to amount the condensation of a natural gas in transmission pipelines. The non isothermal behavior of compressible natural gases mixed with hydrogen in pipelines was analyzed by Uilhoorn [22] using the Predictive Soave–Redlich–Kwong (PSRK) EoS [23]. Absorption properties of hydrogen including fugacity and compressibility factor of hydrogen were evaluated by Zhou and Zhou [24] using the Redlich–Kwong–Soave (RKS) [25] and Benedict–Web–Rubin (BWR) EoSs [26]. The RKS EoS was used at 30 bar and high temperatures by Aktaş et al. [27] in thermodynamic analysis of steam assisted conversion of bio-oil components to synthesis gases. Zheng et al. [28] and Yang and Huber [29] used NIST data base equation [30] for analyzing processes containing hydrogen.

Cubic EoSs are simple, robust and predictive to a certain extent [31,32]. Consequently, cubic EoSs are very popular. However, there are plenty of cubic EoSs in the literature. Excellent reviews of cubic EoSs can be found elsewhere [33–36]. Newly developed EoSs are also emerging. Thus, the accuracy of an EoS must be tested appropriately before using it in a specific application.

Cubic EoSs are generally used to correlate the vapor pressures (and sometimes the liquid densities) of pure substances and predict other properties along the vapor–liquid

coexistence curve. Far from the critical point, the accuracies of cubic EoSs in predicting the supercritical properties are not clear; however.

In this work we evaluate the accuracies of eleven cubic EoSs in predicting the saturated and supercritical properties of hydrogen for wide ranges of temperature and pressure. For accurate prediction of hydrogen properties, the alpha functions and PVT relationships of EoSs are equally important. Therefore, in addition to the widely used RKS EoS [25], PR EoS [16] and Patel–Teja (PT) EoS [37], some of the most successful variants of these EoSs are used in this work. A newly developed EoS [38] is also evaluated. However, we do not use theoretical EoS in our evaluations as these EoSs are often used for chain, associating molecules and electrolytes. Then, we attempt to predict the saturated properties of hydrogen (vapor pressure, saturated liquid and vapor densities, enthalpies and entropies of vaporization, and saturated liquid and vapor heat capacities at constant pressure) using the eleven EoSs. Afterwards, the second virial coefficients, compressibility factors, enthalpies, entropies, constant pressure heat capacities, speeds of sound, Joule–Thomson coefficients and inversion curves of hydrogen are predicted for wide ranges of temperature and pressure. The accuracies of the EoSs in predicting the properties of hydrogen are compared with experimental data, recommended values of thermodynamic tables and also with each other. The best EoSs among the others for predicting hydrogen properties are introduced and their accuracies are reported.

2. The EoSs

With few exceptions, cubic EoSs are mainly similar to the van der Waals EoS [39]. Cubic EoSs differ in pressure–volume–temperature (PVT) relationships, temperature dependence term for the attractive parameter and sometimes in molar co-volume parameter. Temperature dependence term for molar

Table 1 – The PVT relationships for the EoSs.

EoS	PVT	Parameters
RKS	$P = \frac{RT}{v-b} - \frac{a_c \alpha(T_r)}{v(v+b)}$	$b = 0.08664 \frac{RT_c}{P_c}, a_c = 0.42748 \frac{R^2 T_c^2}{P_c}$
RKMC		
RKSW		
PR	$P = \frac{RT}{v-b} - \frac{a_c \alpha(T_r)}{v(v+b) + b(v-b)}$	$b = 0.07780 \frac{RT_c}{P_c}, a_c = 0.45723 \frac{R^2 T_c^2}{P_c}$
PRMC		
PRSV		
PRTCC		
PRGGPR		
PT	$P = \frac{RT}{v-b} - \frac{a_c \alpha(T_r)}{v(v+b) + c(v-b)}$	$\zeta_c = 0.329032 - 0.076799\omega + 0.0211947\omega^2,$ $\Omega_b^3 + (2 - 3\zeta_c)\Omega_b^2 + 3\zeta_c^2\Omega_b - \zeta_c^3 = 0,$ $\Omega_c = 1 - 3\zeta_c, \Omega_a = 3\zeta_c^2 + 3(1 - 2\zeta_c)\Omega_b + \Omega_b^2 + 1 - 3\zeta_c,$ $b = \Omega_b \frac{RT_c}{P_c}, c = \Omega_c \frac{RT_c}{P_c}, a_c = \Omega_a \frac{R^2 T_c^2}{P_c}$
PTV		$b = (0.02207 + 0.20868 Z_c) \frac{RT_c}{P_c}, c = (0.57765 - 1.87080 Z_c) \frac{RT_c}{P_c},$ $a_c = (0.66121 - 0.76105 Z_c) \frac{R^2 T_c^2}{P_c}$
NB	$P = \frac{RT}{v-b} - \frac{a_c \alpha(T_r)}{(v+\beta b)^2}$	$b = 0.079246 \frac{RT_c}{P_c}, \beta = 1/\sqrt{3}, a_c = 0.421875 \frac{R^2 T_c^2}{P_c}$

Table 2 – Alpha functions for the EoSs.

EoS	$\alpha(T_r)$
RKS	$\alpha = [1 + m(1 - \sqrt{T_r})]^2, m = 0.480 + 1.574\omega - 0.176\omega^2.$
RKMC, PRMC	$\alpha = \begin{cases} [1 + c_1(1 - \sqrt{T_r}) + c_2(1 - \sqrt{T_r})^2 + c_3(1 - \sqrt{T_r})^3]^2 & T_r \leq 1 \\ [1 + c_1(1 - \sqrt{T_r})]^2 & T_r > 1 \end{cases}$
RKSW	$\alpha = \begin{cases} [1 + m(1 - \sqrt{T_r})]^2 & T_r \leq 1 \\ \frac{b_1 + \frac{b_2}{T_r} + \frac{b_3}{T_r^2}}{T_r} & T_r > 1 \end{cases}, m = 0.480 + 1.574\omega - 0.176\omega^2, b_1 = 0.25(m^2 - 11m + 12), b_2 = 0.5(-m^2 + 9m - 6), b_3 = 0.25(m^2 - 7m + 4).$
PR	$\alpha = [1 + m(1 - \sqrt{T_r})]^2, m = 0.3746 + 1.5423\omega - 0.2699\omega^2.$
PRSV	$\alpha = [1 + m(1 - \sqrt{T_r})]^2, k_0 = 0.378893 + 1.4897153\omega - 0.1713184\omega^2 + 0.0196554\omega^3, m = k_0 + k_1(1 + \sqrt{T_r})(0.7 - T_r).$
PRTCC	$\alpha_0 = T_r^{-0.171813} \exp[0.125283(1 - T_r^{1.77634})], \alpha_0 = T_r^{-0.607352} \exp[0.511614(1 - T_r^{2.0517})], \alpha = \alpha_0 + \omega(\alpha_1 - \alpha_0)$
PRGGPR	$\alpha = \exp[(2 + 0.836T_r)(1 - T_r^{0.134+0.508\omega-0.0467\omega^2})]$
PT	$\alpha = [1 + m(1 - \sqrt{T_r})]^2, m = 0.452413 + 1.30982\omega - 0.295937\omega^2.$
PTV	$\alpha = [1 + m(1 - \sqrt{T_r})]^2, m = 0.46283 + 3.5823Z_c\omega - 0.295937(Z_c\omega)^2.$
NB	$\alpha = \begin{cases} [1 + m(1 - \sqrt{T_r})]^2 & T_r \leq 1 \\ \frac{b_1 + \frac{b_2}{T_r} + \frac{b_3}{T_r^2}}{T_r} & T_r > 1 \end{cases}, m = 0.4857 + 1.6308\omega - 0.2089\omega^2, b_1 = 0.25(m^2 - 11m + 12), b_2 = 0.5(-m^2 + 9m - 6), b_3 = 0.25(m^2 - 7m + 4).$

Table 3 – The critical properties, molecular weight and acentric factor of hydrogen for calculating the EoSs parameters (except for the PRSV EoS) [49].

M_w	2.016
$T_c(K)$	33.2
$P_c(MPa)$	1.3
Z_c	0.306
ω	-0.218

co-volume parameter is often improving an EoS in predicting liquid densities; however, it may lead to isotherms crossover at high pressures [40]. Therefore, temperature dependence for molar co-volume parameter is not recommended [40]. In this work we employ two- and three-parameter EoSs with constant molar co-volume parameters.

2.1. Two-parameter EoSs

2.1.1. Redlich–Kwong family

Redlich and Kwong (RK) [41] introduced one of the most successful PVT relationships. Later, Soave [25] modified the RK EoS successfully. The RKS EoS has found widespread applications in chemical and petroleum industries. Thus, the RKS EoS is evaluated in this work. Tables 1 and 2 give, respectively, the PVT relationship and temperature dependence term, also called alpha function, for the RKS EoS. In Tables 1 and 2, one can also find two other successful variants of the RK family, i.e., the modified Redlich–Kwong EoS by Mathias and Copeman (RKMC) [42] and by Nasrifar and Bolland (RKSW) [43]. The RKMC is more accurate than the RKS EoS in correlating the vapor pressure of pure compounds. However, the temperature dependence of the RKMC EoS is more elaborate than the RKS EoS. The RKMC EoS also needs three specific adjustable parameters for each compound. The RKSW EoS has recently been developed to improve the incorrect behavior of the RKS EoS at high temperatures. The RKSW EoS employs the square-well (SW) potential to account for the intermolecular interactions of supercritical fluids. As shown by Nasrifar and Bolland [43], the RKSW EoS is often more accurate than the RKS EoS in predicting the volumetric properties of pure substances. Both the RKMC and RKSW EoSs are used in this work.

2.1.2. Peng–Robinson family

As the liquid densities predicted by the RK family EoSs were poor, Peng and Robinson (PR) [16] developed a more accurate PVT relationship than that of the RK EoS. Consequently, the PR EoS is the most widely used EoS in industry. Many

Table 4 – The classical critical properties and acentric factor of hydrogen for calculating the parameters of the PRSV EoS. The adjustable parameter of the PRSV EoS is also given [50].

$T_c(K)$	43.6
$P_c(MPa)$	20.47
Z_c	0.242
ω	0
κ_1	0

Table 5 – Mathias and Copeman coefficients for the alpha functions of the RKMC and PRMC EoSs (the parameters of the RKMC EoS are from Ref. [52] and for the PRMC EoS were determined in this work).

	RKMC	PRMC
c_1	0.2053	0.0675
c_2	−0.4241	−0.0554
c_3	−0.1032	−0.5662

Table 6 – Ideal gas state heat capacity of hydrogen [53].

$c_p^o (J\ kmol^{-1}K^{-1}) = d_1 + d_2 \left[\frac{d_3}{T(K)} \sin h \left(\frac{d_3}{T(K)} \right) \right]^2 + d_4 \left[\frac{d_5}{T(K)} \cos h \left(\frac{d_5}{T(K)} \right) \right]^2$	
d_1	27,620
d_2	9560
d_3	2466
d_4	3760
d_5	567.6

modifications on the PR EoS have been made to improve the predictions of the PR EoS along the coexistence curves [42] and at supercritical conditions [44,46]. The modifications made by Mathias and Copeman (PRMC) [42], Twu et al. (PRTCC) [44], Stryjek and Vera (PRSV) [45], and Gasem et al. (PRGGPR) [46] are the most successful ones among the others. These EoSs have the same PVT relationship but different alpha functions. Tables 1 and 2 provide the PVT relationship and alpha functions of these EoSs, respectively. The PR, PRMC, PRTCC, PRSV and PRGGPR EoSs are also evaluated in this work.

The last two-parameter equation of state that is chosen for predicting hydrogen properties was developed by Nasrifar and Bolland (NB) [38]. The NB EoS is not classified among the RK or PR families. It has different PVT relationship. The NB EoS was initially developed for predicting natural gas properties. However, as shown recently [32], the NB EoS is quite accurate for predicting the saturated vapor volume of many classes of substances. The NB EoS takes the advantages of the SW

Table 8 – Accuracies of the EoSs in predicting the supercritical properties of hydrogen.

EoS	%AAD ^a						
	B_2^b	z^c	h^d	s^d	c_p^d	θ^e	μ_{JT}^f
RKS	29.17	0.66	0.33	0.13	0.64	3.17	21.99
RKMC	13.00	0.82	0.16	0.10	0.67	3.43	4.26
RKSW	12.08	1.05	0.26	0.08	0.78	4.03	16.11
PR	81.60	2.39	1.13	0.14	0.66	3.27	73.66
PRMC	69.78	1.95	0.95	0.14	0.66	2.93	60.16
PRSV	63.61	3.58	0.91	0.17	1.49	2.78	47.39
PRTCC	28.78	0.87	0.47	0.15	1.01	1.50	40.02
PRGGPR	66.43	1.75	0.88	0.17	0.75	2.97	58.46
PT	14.13	1.47	0.20	0.46	0.60	3.66	9.71
PTV	6.49	1.02	0.23	0.44	0.78	3.63	11.04
NB	8.52	0.41	0.15	0.42	0.78	2.98	8.56

^a %AAD = $(100/n) \sum_i^n |\text{calc}_i - \text{expl}_i| / \text{expl}_i$

^b Experimental data from Ref. [55]; T Range: 14–400 K; n = 15.

^c Experimental data from Ref. [55]; T Range: 50–1500 K; P Range: 0.1–100 MPa; n = 393.

^d Experimental data from Ref. [55]; T Range: 160–1500 K; P Range: 0.1–100 MPa; n = 324.

^e Experimental data from Ref. [55]; T Range: 50–1500 K; P Range: 0.1–100 MPa; n = 356.

^f Experimental data from Ref. [30]; T Range: 160–900 K; P Range: 0.1–100 MPa; n = 130.

potential for predicting the supercritical properties of pure compounds. The PVT and alpha function for the NB EoS are also given in Tables 1 and 2.

2.2. Three-parameter EoSs

Predictions of liquid densities deteriorate near the critical points of pure substances using the RK and PR family EoSs. Patel and Teja (PT) [37] recognized that incorporating a conventional critical compressibility term would circumvent this shortcoming. Consequently, the PT EoS is one of the best cubic EoSs available in the literature. However, it has three-parameters (a, b, c) and a little more complex than the

Table 7 – Accuracies of the EoSs in predicting the saturated properties of hydrogen (T Range: 14–32 K, n = 22, experimental data from Ref. [55]).

EoS	%AAD ^a						
	p^{sat}	v^l	v^v	Δh^{vap}	Δs^{vap}	c_p^l	c_p^v
RKS	6.44	8.04	8.82	7.68	7.67	86.25	17.57
RKMC	1.31	7.52	1.74	1.88	2.18	25.38	16.45
RKSW	6.44	8.04	8.82	7.68	7.67	86.25	17.57
PR	3.36	14.45	5.65	5.96	6.32	73.76	17.83
PRMC	0.88	15.41	2.09	2.64	2.88	28.88	17.19
PRSV	–	–	–	–	–	–	–
PRTCC	0.91	14.88	1.44	2.06	2.28	36.33	17.07
PRGGPR	6.62	15.01	9.99	7.03	7.04	80.82	17.54
PT	26.78	7.06	24.16	8.94	9.57	121.98	34.08
PTV	22.19	9.44	44.96	19.49	19.23	109.44	16.32
NB	4.91	10.35	6.39	7.10	7.28	84.16	17.79

^a %AAD = $(100/n) \sum_i^n |\text{calc}_i - \text{expl}_i| / \text{expl}_i$

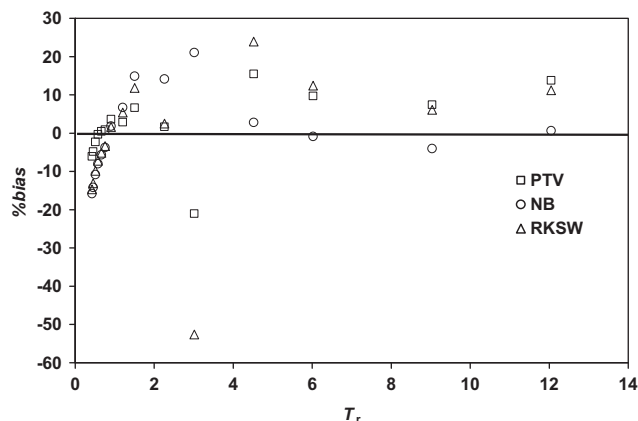


Fig. 1 – Deviation plot for predicting the second virial coefficients of hydrogen ($\%bias = 100(B_2^{calc} - B_2^{expl})/B_2^{expl}$ and experimental data from Ref. [54]).

RK and PR family EoSs. Valderrama (PTV) [47] introduced a simplified but more accurate version of the PT EoS. The PTV EoS has been applied successfully in gas hydrate applications [48]. Tables 1 and 2 provide the PVT and alpha functions for the PT and PTV EoSs, respectively. These two EoSs are also used in predicting hydrogen properties.

3. Results and discussion

The molecular weight, acentric factor and critical properties of hydrogen are required to calculate the EoS parameters and consequently the thermodynamic properties of hydrogen. The critical properties and constants are given in Table 3. For the PRSV EoS; however, we used the classical critical properties reported in DECHEMA Chemistry Data Series [50] with the parameter κ_1 set to zero. These values are given in Table 4. The classical critical values given in Table 4 suggest a compressibility factor of 0.242 for hydrogen. This value is gained at high temperatures where hydrogen shows classical behavior [51]. The use of these critical properties and constants was

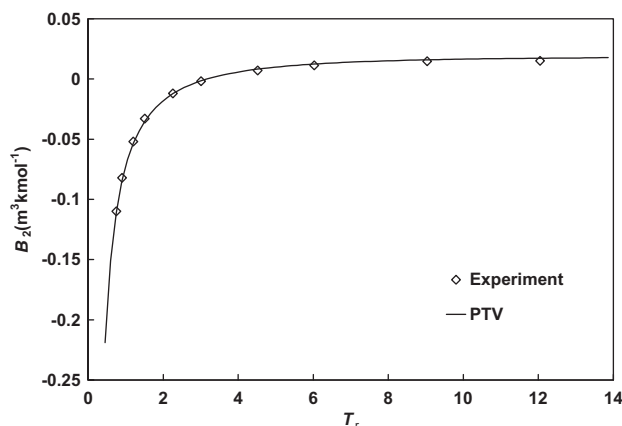


Fig. 2 – Recommended [54] and predicted second virial coefficients of hydrogen with the PTV EoS.

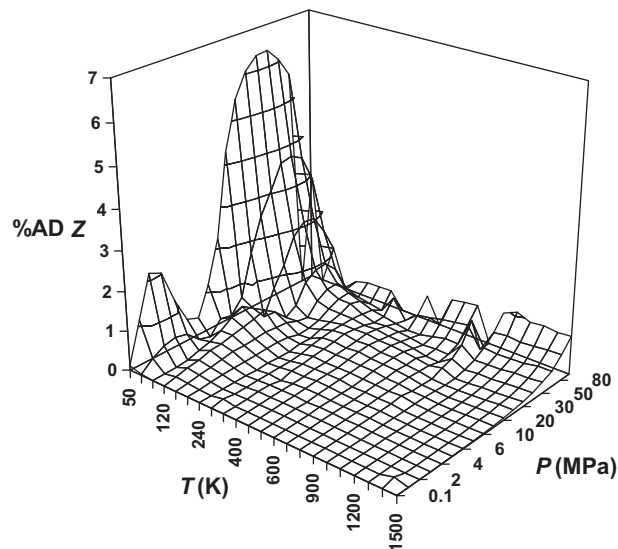


Fig. 3 – Deviation plot for predicting the compressibility factors of hydrogen with the NB EoS ($\%bias = 100|z^{calc} - z^{expl}|/z^{expl}$ and experimental data from Ref. [55]).

suggested by Huang et al. [13], where the authors successfully correlated the vapor–liquid equilibria of hydrogen + hydrocarbon systems. For the RKMC and PRMC EoSs, the coefficients c_1 , c_2 and c_3 of the Mathias and Copeman [42] alpha function are required. We used the parameters provided by Nasrifar and Moshfeghian [52] for the RKMC EoS. For the PRMC EoS, we determined the coefficients by fitting the PRMC EoS to the hydrogen vapor pressures. These parameters are given in Table 5. Ideal gas state heat capacity for hydrogen [53], where it is used in predicting thermal and related properties of hydrogen, can be found in Table 6.

The eleven EoSs were used to predict the saturated properties of hydrogen. The results were collected in Table 7. As can be seen, the quality of the results is poor in predicting the saturated liquid and vapor heat capacities of hydrogen at constant pressure. However, for predicting the vapor pressure, the

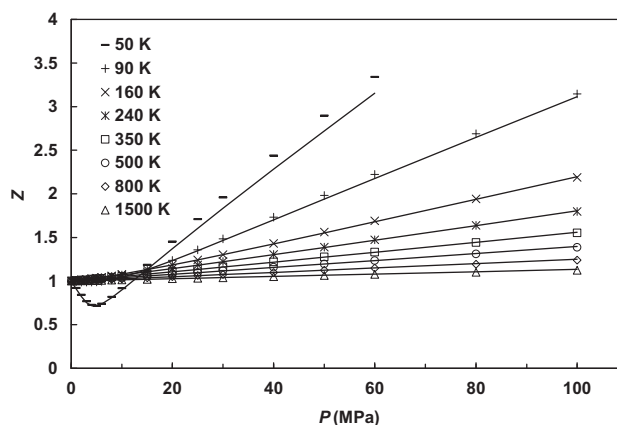


Fig. 4 – Experimental [55] and predicted compressibility factors of hydrogen with the NB EoS.

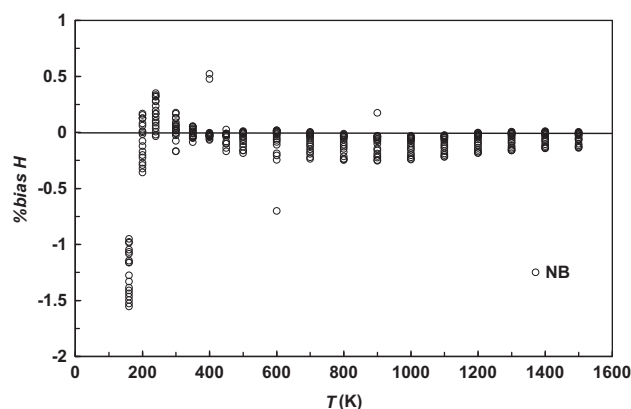


Fig. 5 – Deviation plot for predicting the enthalpies of hydrogen. The NB EoS was used for the predictions, $\%bias = 100(h^{calc} - h^{expl})/h^{expl}$ and experimental data from Ref. [55].

saturated vapor volume, and vaporization enthalpies and entropies of hydrogen, the RKMC and PRTCC EoSs are quite accurate. For predicting the saturated liquid volumes, the RKMC EoS is considerably better than the PRTCC EoS. In general, the RKMC and PRTCC EoSs are recommended for predicting the saturated properties of hydrogen among the others. Table 7 also indicates that the PRSV EoS is not competent in predicting the properties of saturated hydrogen. Although the PRSV EoS was successfully used in mixture equilibrium calculations [13], it fails in calculating the saturated properties of pure hydrogen. In other words, the accuracy of an EoS in predicting the properties of saturated hydrogen is sufficient but not necessary for predicting mixture equilibrium properties. Briefly, the two-parameter EoSs are more accurate than the three-parameter EoSs in predicting saturated properties of hydrogen. Both the RK and PR families are accurate to the same extent. In the RK family, the RKMC is the best among the others. In the PR family, PRTCC is the best among the others. The accuracy of the NB EoS is also similar to the RKS EoS.

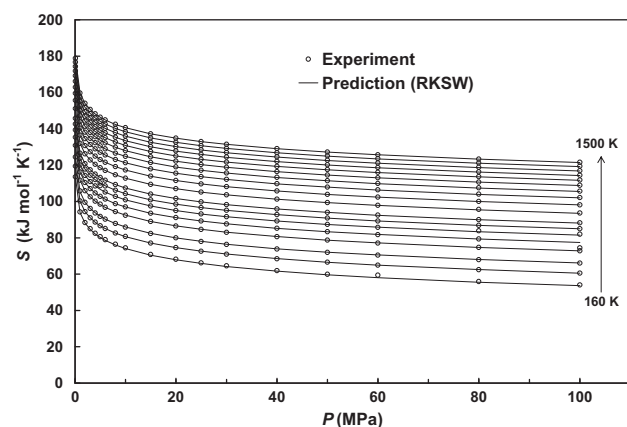


Fig. 6 – Experimental [55] and predicted entropies of hydrogen with the RKSW EoS.

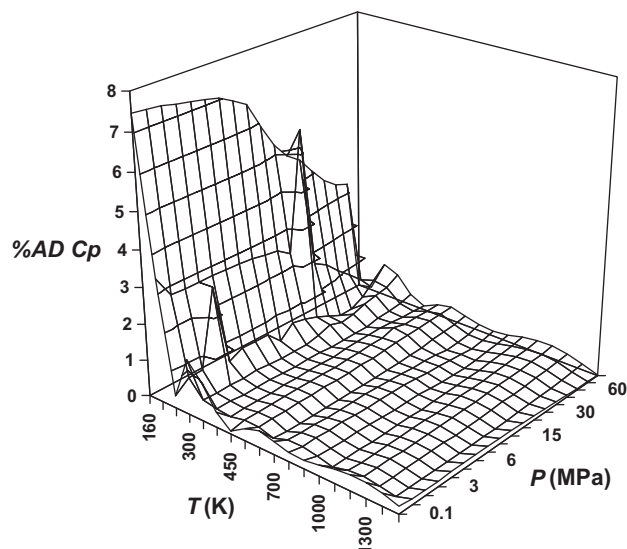


Fig. 7 – Deviation plot for predicting the constant pressure heat capacities of hydrogen with the PT EoS ($\%bias = 100|c_p^{calc} - c_p^{expl}|/c_p^{expl}$ and experimental data from Ref. [55]).

Table 8 presents the accuracy of the eleven EoSs in predicting the supercritical properties of hydrogen for wide ranges of temperature and pressure. Table 8 indicates that the second virial coefficient of hydrogen expressed by

$$B_2(T) = b - \frac{a_c \alpha(T_r)}{RT} \quad (1)$$

is best predicted by the PTV, NB and RKSW EoSs, respectively. In Eq. (1), b is the molar co-volume parameter, a_c is the attractive parameter at the critical point, $\alpha(T_r)$ is the alpha function, R is the universal gas constant, T is the temperature and T_r is the reduced temperature (T/T_c). Fig. 1 is a deviation plot for predicting the second virial coefficient of hydrogen using the PTV, NB and RKSW EoSs as a function of reduced temperature (T/T_c). Fig. 1 shows that at small values of

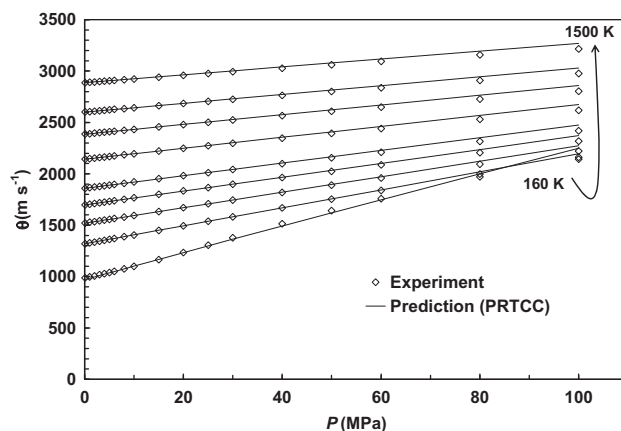


Fig. 8 – Experimental [55] and predicted speeds of sound data of hydrogen with the PRTCC EoS.

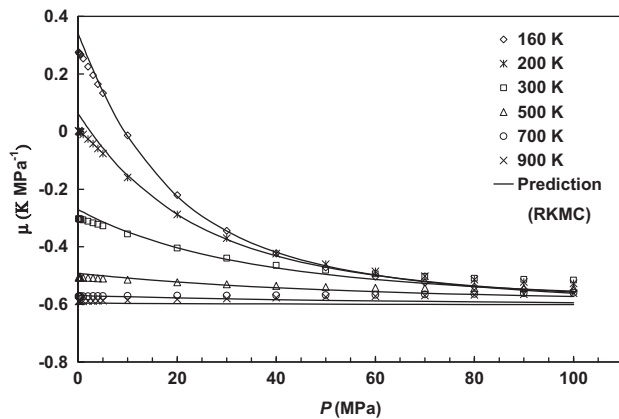


Fig. 9 – Recommended [30] and predicted Joule-Thomson coefficients of hydrogen with the RKMC EoS.

reduced temperature ($T_r \leq 4$), the PTV EoS is comparatively more accurate than the NB and RKS EoSs. However, for larger values of reduced temperature ($4 \leq T_r \leq 12$) the NB EoS is the most accurate EoS in predicting the second virial coefficient of hydrogen. There are also accurate corresponding state correlations for prediction of the second virial coefficients of normal fluids [49,51]. However, hydrogen is a quantum fluid and these correlations are inapplicable. In Fig. 2, it is seen that the PTV EoS markedly predicts the recommended second virial coefficients [54].

Table 8 also indicates that the compressibility factors of hydrogen for temperature range from 50 K to 1500 K and pressure range from 0.1 MPa to 100 MPa are accurately predicted by the NB, RKS and RKMC EoSs, respectively. Fig. 3 is a deviation plot for predicting the compressibility factors of hydrogen with the NB EoS. Fig. 3 shows that at small temperatures ($T \leq 160$ K) errors are large. This can be attributed to the quantum behavior of hydrogen molecules. For temperatures larger than 160 K, the errors are small, however. Fig. 4 depicts the experimental [55] and predicted compressibility factors of hydrogen using the NB EoS. Clearly, at 50 K and at high pressures, the NB EoS has the least accuracy. However, for the whole range as indicated in Table 8, the NB EoS is the most accurate cubic EoS among the others. Indicated in Table 8, the PR family is less accurate than the RK and PT families in predicting hydrogen compressibility factors.

The accuracies of the EoSs in predicting the enthalpies of hydrogen are also given in Table 8. In predicting hydrogen enthalpies, the RK and PT families have similar accuracies. The accuracies of the PR family are a little worse. However, the NB, RKMC and PT EoSs are the most accurate EoSs, respectively. Deviation plot for the NB EoS in predicting the enthalpies of hydrogen as a function of temperature is illustrated in Fig. 5. Shown in Fig. 5, the NB EoS is reasonably accurate at temperatures smaller than 200 K. At larger temperatures; however, the NB EoS quite accurately predicts the enthalpies of hydrogen. Therefore, it seems that at temperatures equal or less than 160 K, hydrogen shows non-ideality due to quantum behavior. At high temperatures, this effect vanishes. In a word, the temperature 160 K is the threshold value for the classical behavior of hydrogen.

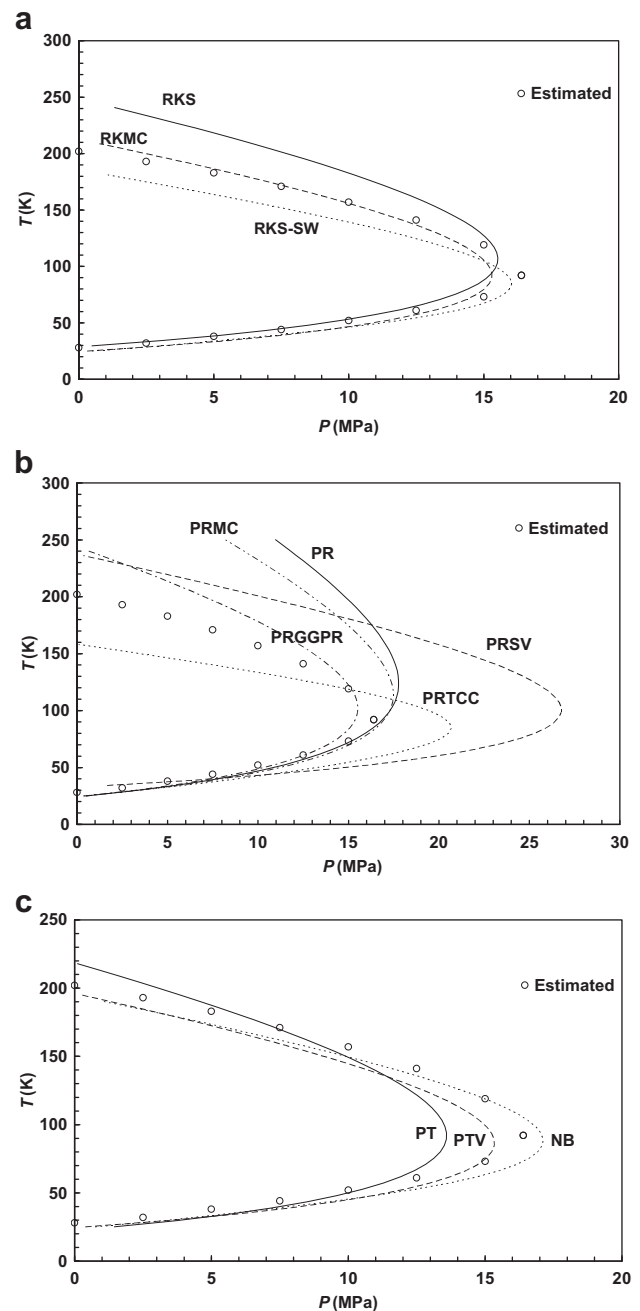


Fig. 10 – Estimated [53] and predicted Joule-Thomson inversion curves with (a) the RK family EoSs, (b) the PR family EoSs, (c) the PT, PTV and NB EoSs.

In predicting the entropies of hydrogen as presented in Table 8, the RKS, RKMC and RKS EoSs are the most accurate EoSs. In fact, the RK family is the most accurate EoSs while the PT family the least. As can be seen, the other EoSs are also accurate. Fig. 6 demonstrates the experimental and predicted entropies of hydrogen using the RKS EoS. Clearly, the agreement between the predictions and experimental values are quite good.

Table 8 also presents the accuracy of the EoSs in predicting constant pressure heat capacities of hydrogen. The constant pressure heat capacity of hydrogen is calculated from

Table 9 – Summary of the best EoSs for predicting the thermophysical properties of hydrogen (pressure ranges from 0.1 to 100 MPa).

Property(s)	EoSs	n	T. Range (K)	%AAD ^a
Saturated properties (overall) ^b	RKMC, PRTCC	154	14–32	8.06, 10.71 ^c
Supercritical properties (overall)	RKMC, NB, PTV	1866	160–1500	1.38, 1.55, 1.98 ^d
Second virial coefficient	PTV, NB, RKSW	15	14–400	6.49, 8.52, 12.08
Compressibility factor	NB, RKS, RKMC	393	50–1500	0.41, 0.66, 0.82
Enthalpy	NB, RKMC, PT	324	160–1500	0.15, 0.16, 0.20
Entropy	RKSW, RKMC, RKS	324	160–1500	0.08, 0.10, 0.13
Heat Capacity	PT, RKS, PR	324	160–1500	0.60, 0.64, 0.66
Speed of Sound	PRTCC, PRSV, PRGGPR	356	50–1500	1.50, 2.78, 2.97
Joule–Thomson coefficient	RKMC, NB, PT	130	160–900	4.26, 8.56, 9.71

a %AAD = $(100/n) \sum_i^n |\text{calc}_i - \text{expl}_i| / \text{expl}_i$

b Applied pressure is saturated pressure.

c Average of %AAD for all calculated saturated properties.

d Average of %AAD for all calculated supercritical properties.

$$c_p = c_p^\circ(T) + T \int_{\infty}^v \left(\frac{\partial^2 p}{\partial T^2} \right)_v dv - \frac{T[(\partial p / \partial T)_v]^2}{(\partial p / \partial v)_T} - R \quad (2)$$

where $c_p(T)$ is the constant pressure heat capacity, $c_p^\circ(T)$ is the ideal gas state heat capacity as given for hydrogen in Table 6 and v is the volume. In Table 8, one can see that all EoSs in all families are accurate in predicting the constant pressure heat capacities of hydrogen. However, the most accurate EoSs are the PT, RKS and PR EoSs, respectively. In working with the PT EoS, one should note that with the critical properties given in Table 3, the parameter c of the PT EoS (see Table 1) will find a negative value. Because a negative value is unrealistic, we have set $c = 0$ in all calculations throughout the paper. Fig. 7 demonstrates a deviation plot for the PT EoS in predicting the constant pressure heat capacities of hydrogen. The PT EoS presents inaccuracies at 160 K but errors decrease dramatically at high temperatures.

For predicting speeds of sound in hydrogen, the PRTCC, PRSV and PRGGPR EoSs are the best among the others (see Table 8). In fact, the PR family is quite accurate when compared to experimental data and the other EoSs. The speed of sound is calculated from

$$\theta^2 = -\gamma \frac{v^2}{M_w} \left(\frac{\partial p}{\partial v} \right)_T \quad (3)$$

where θ is the sound velocity, M_w is the molecular weight, γ is the heat capacity ratio (c_p/c_v) and c_v is the constant volume heat capacity calculated by

$$c_v = c_v^\circ(T) + T \int_{\infty}^v \left(\frac{\partial^2 p}{\partial T^2} \right)_v dv \quad (4)$$

In Eq. (4), c_v° is the ideal gas state heat capacity at constant volume ($c_p^\circ - R$). Fig. 8 shows that the PRTCC quite accurately predicts the speeds of sound data from 0.1 MPa to 100 MPa and temperatures from 160 K to 1500 K.

Joule–Thomson coefficient is expressed by

$$\mu_{JT} = -\frac{1}{c_p} \left[v + T \frac{(\partial p / \partial T)_v}{(\partial p / \partial v)_T} \right] \quad (5)$$

where μ_{JT} is the Joule–Thomson coefficient. Accuracies of the EoSs in predicting the Joule–Thomson coefficients of hydrogen

are given in Table 8. Clearly, the PR family EoSs cannot predict the Joule–Thomson coefficients of hydrogen. The PT family is indeed better than the RK family. Table 8, however, clarifies that the predictions from the RKMC EoS are in best agreement with the recommended values of NIST [30]. The percent average absolute deviation (%AAD) was found to be 4.26 for the RKMC EoS. The next best EoSs in predicting the Joule–Thomson coefficients of hydrogen were the NB and PT EoSs, respectively. The %AAD for the NB and PT EoSs were found to be 8.56 and 9.71, respectively. Fig. 9 exhibits the Joule–Thomson coefficients as a function of pressure with temperature as a second parameter. The agreement is satisfactory between the predictions with the RKMC EoS and the recommended values [30]. However, at high temperatures (larger than 500 K) the agreement is less satisfactory, especially at high pressures.

The ability of an EoS to predict Joule–Thomson inversion curves is a stringent test of the EoS accuracy. Joule–Thomson inversion curves describe the attractive part of the EoSs. Inversion curves are the loci of conditions where μ_{JT} vanishes. The Joule–Thomson inversion curve is then expressed by

$$\mu_{JT} = 0 \quad (6)$$

Fig. 10a, b and c show the inversion curves for hydrogen using different families of EoSs. Shown in Fig. 10a, the RKMC EoS markedly predicts the inversion curve. The two other members of the RK family; however, rather accurately predict the lower branch of the inversion curves. These two EoSs fail to predict the upper branch. In Fig. 10b, one can see that the PR family is not competent for predicting the inversion curve of hydrogen. Nevertheless, the PR, PRMC and PRGGPR EoSs satisfactorily predict the lower branch. Finally, Fig. 10c depicts the ability of the PT, PTV and NB EoSs in predicting the inversion loci. The NB EoS is a little superior compared to the PT and PTV EoSs. The PTV is also rather accurate. Fig. 10a, b and c indicate that the RKMC EoS is superior compared to the other EoSs. The next best EoSs are the NB and PTV EoSs, respectively.

Overall, in predicting the supercritical properties of hydrogen, the two-parameter EoSs are as good as or more accurate than the three-parameter EoSs. Among the two-parameter EoSs, the RK family is superior while the PR family inferior. Among the RK family EoSs, the best predictive EoS is the RKMC EoS. While in the PR family, the best EoS in predicting

hydrogen properties is the PRTCC EoS. The two-parameter NB EoS is a little better than the three-parameter PTV EoS.

4. Conclusions

Calculations performed in this study made it clear that for predicting the properties of hydrogen for the whole range (saturated and supercritical conditions) the RKMC EoS is accurate and robust. Therefore, the RKMC EoS is recommended for most applications connected with hydrogen. For predicting hydrogen properties at temperatures greater than 200 K, almost all EoSs can be used with comparable accuracy. However, the best EoSs for predicting different hydrogen properties are summarized in Table 9. Table 9 is a quick reference for choosing the most appropriate EoS for an application. It is worth stressing that for accurate prediction of hydrogen properties, the differences in alpha functions and PVT relationships of the EoSs make them behave differently.

Nomenclature

a	attractive parameter, $\text{MPa m}^6 \text{ k mol}^{-2}$
b	molar co-volume, $\text{m}^3 \text{ k mol}^{-1}$
B_2	second virial coefficient, $\text{m}^3 \text{ k mol}^{-1}$
c	the third parameters of PT, PTV
c_1, c_2, c_3	coefficients of the Mathias and Copeman alpha function (see Table 5)
c_p	heat capacity at constant pressure, $\text{kJ k mol}^{-1} \text{ K}^{-1}$
c_v	heat capacity at constant volume, $\text{kJ k mol}^{-1} \text{ K}^{-1}$
calc	calculated value
d_1, d_2, d_3, d_4, d_5	constants of standard state heat capacity (see Table 6)
expl	experimental value
h	enthalpy, kJ kmol^{-1}
i	dummy index
m	parameter defined in the α function of the EoSs (see Table 2)
n	number of points used in evaluations
M_w	molecular weight
p	pressure, MPa
R	universal gas constant, $8.314 \times 10^{-3} \text{ MPa m}^3 \text{ kmol}^{-1} \text{ K}^{-1}$
s	entropy, $\text{kJ kmol}^{-1} \text{ K}^{-1}$
T	temperature, K
v	molar volume, $\text{m}^3 \text{ k mol}^{-1}$
z	compressibility factor

Greek letters

α	temperature dependence function (alpha function)
γ	specific heat ratio
κ_0	generalized parameter of the PRSV EoS
κ_1	adjustable parameter of the PRSV EoS ($\kappa_1 = 0$ for hydrogen)
μ	Joule–Thomson coefficient, K MPa^{-1}
θ	speed of sound, m s^{-1}
ω	acentric factor
Ω	parameter defined in the PT EoS

ζ_c ad-hoc critical compressibility factor used in the PT EoS

Subscripts/superscripts

a, b, c	attractive, molar co-volume and the third parameter of the PT EoS
c	critical property
l	liquid
p	pressure
r	reduced property
sat	saturated property
v	vapor/volume
vap	vaporization
JT	Joule–Thomson
◦	standard state

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